

UNIT III

INTEGRAL EQUATIONS

Verification of Solutions & the Method of Successive Approximations

Section 1: Solution of Integral Equations

Section 2: Method of Successive Approximations

Section 3: Corrections and Summary

M.Tech. (ASE) — Class Notes, Fully Worked

Every step of every derivation shown in full

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1 Solution of Integral Equations

1.1 Basic Definitions

Integral Equation (I.E.)

An equation in which the **unknown function** appears under an **integral sign** is called an **integral equation**. For example,

$$y(x) + \int_0^x y(t) dt = 0.$$

Integral equations are mainly used to solve **boundary value problems** arising in mechanics and mathematical physics.

General linear integral equation

$$y(x) = F(x) + \lambda \int_a^b K(x, t) y(t) dt$$

- $F(x)$ and $K(x, t)$ are **known** functions; $K(x, t)$ is called the **kernel**.
- $y(x)$ is the **unknown** function to be determined.
- λ is a parameter.

Fredholm and Volterra equations

Fredholm integral equation: both limits a and b are **constants**,

$$y(x) = F(x) + \lambda \int_a^b K(x, t) y(t) dt.$$

Volterra integral equation: a is a **constant** and the upper limit is the **variable** x ,

$$y(x) = F(x) + \lambda \int_a^x K(x, t) y(t) dt.$$

1.2 Solution of an Integral Equation

Solution

A function $y(x)$ is a **solution** of a given integral equation if, on substituting it into the equation, the equation is **identically satisfied**.

Working rule

1. Substitute the proposed y into **one** side (usually the side containing the integral).
2. Evaluate the integral **completely**.
3. Simplify and compare with the other side.
4. If LHS = RHS, then $y(x)$ is a solution.

1.3 Problem 1: $y = 2 - x$ **Problem 1**

Show that $y(x) = 2 - x$ is a solution of the integral equation

$$\int_0^x e^{x-t} y(t) dt = e^x + x - 1. \quad (1)$$

Plan. Put $y(t) = 2 - t$ into the left-hand side of (1), evaluate the integral step by step, and check that the result equals $e^x + x - 1$.

Step 1: Substitute and separate

Let $y(x) = 2 - x$, so that $y(t) = 2 - t$. Then

$$\begin{aligned} \text{LHS} &= \int_0^x e^{x-t} (2 - t) dt = \int_0^x e^x e^{-t} (2 - t) dt && \text{since } e^{x-t} = e^x e^{-t} \\ &= e^x \int_0^x (2 - t) e^{-t} dt = e^x I, && e^x \text{ is constant w.r.t. } t \end{aligned}$$

where $I = \int_0^x (2 - t) e^{-t} dt$.

Step 2: Bernoulli's formula**Bernoulli's (generalised integration by parts) formula**

$$\int u v dt = u v_1 - u' v_2 + u'' v_3 - \dots$$

where $v_1, v_2, v_3, \dots$ are successive integrals of v .

Take $u = 2 - t$ and $v = e^{-t}$:

$$\begin{aligned} u &= 2 - t, & u' &= -1, & u'' &= 0, \\ v_1 &= \int e^{-t} dt = -e^{-t}, & v_2 &= \int v_1 dt = e^{-t}. \end{aligned}$$

Step 3: Antiderivative

The formula terminates because $u'' = 0$:

$$\begin{aligned} \int (2 - t) e^{-t} dt &= (2 - t)(-e^{-t}) - (-1)(e^{-t}) = -(2 - t) e^{-t} + e^{-t} \\ &= e^{-t} [-(2 - t) + 1] = (t - 1) e^{-t}. \end{aligned}$$

Check by differentiating

$$\frac{d}{dt} [(t - 1) e^{-t}] = e^{-t} - (t - 1) e^{-t} = e^{-t} (2 - t) \quad \checkmark$$

Step 4: Apply the limits

$$\begin{aligned} I &= \left[-(2 - t) e^{-t} + e^{-t} \right]_0^x = \underbrace{\left(-(2 - x) e^{-x} + e^{-x} \right)}_{\text{at } t=x} - \underbrace{\left(-(2 - 0) e^0 + e^0 \right)}_{\text{at } t=0} \\ &= -(2 - x) e^{-x} + e^{-x} - (-2 + 1) = -(2 - x) e^{-x} + e^{-x} + 1. \end{aligned}$$

Equivalently, $I = (x - 1) e^{-x} - [(0 - 1) e^0] = (x - 1) e^{-x} + 1$.

Step 5: Multiply back by e^x

$$\begin{aligned}\text{LHS} &= e^x I = e^x \left[-(2-x)e^{-x} + e^{-x} + 1 \right] = -(2-x) \underbrace{e^x e^{-x}}_{=1} + \underbrace{e^x e^{-x}}_{=1} + e^x \\ &= -(2-x) + 1 + e^x = x - 2 + 1 + e^x = x - 1 + e^x.\end{aligned}$$

Conclusion

LHS = $e^x + x - 1$ = RHS. Hence, on substituting $y(x) = 2 - x$ the given equation is satisfied identically:

$$y(x) = 2 - x \text{ is a solution of } \int_0^x e^{x-t} y(t) dt = e^x + x - 1$$

Remark. Because the upper limit is the variable x , this is a **Volterra integral equation** (of the first kind).

1.4 Problem 2: $y = (1 + x^2)^{-3/2}$ **Problem 2**

Show that $y(x) = (1 + x^2)^{-3/2}$ is a solution of the Volterra integral equation

$$y(x) = \frac{1}{1+x^2} - \int_0^x \frac{t}{1+x^2} y(t) dt. \quad (2)$$

Plan. Substitute $y(t) = (1 + t^2)^{-3/2}$ into the RHS of (2) and show that the RHS reduces to $(1 + x^2)^{-3/2}$, i.e. to the LHS.

Step 1: Substitute

$$\text{RHS} = \frac{1}{1+x^2} - \int_0^x \frac{t}{1+x^2} (1+t^2)^{-3/2} dt = \frac{1}{1+x^2} - \frac{1}{1+x^2} \int_0^x \frac{t dt}{(1+t^2)^{3/2}}.$$

The integration variable is t ; the factor $\frac{1}{1+x^2}$ contains only x , so it is a **constant** for this integral and may be taken outside.

Step 2: Substitution $m = 1 + t^2$ **Change of variable**

$$m = 1 + t^2 \implies 2t dt = dm \implies t dt = \frac{1}{2} dm; \quad t = 0 \implies m = 1, \quad t = x \implies m = 1 + x^2.$$

Step 3: Integrate in m

$$\begin{aligned}\text{RHS} &= \frac{1}{1+x^2} - \frac{1}{1+x^2} \cdot \frac{1}{2} \int_1^{1+x^2} m^{-3/2} dm = \frac{1}{1+x^2} - \frac{1}{1+x^2} \cdot \frac{1}{2} \left[\frac{m^{-1/2}}{-\frac{1}{2}} \right]_1^{1+x^2} \\ &= \frac{1}{1+x^2} - \frac{1}{1+x^2} \cdot \frac{1}{2} \cdot (-2) \left[m^{-1/2} \right]_1^{1+x^2} = \frac{1}{1+x^2} + \frac{1}{1+x^2} \left[m^{-1/2} \right]_1^{1+x^2}.\end{aligned}$$

Note on signs: the two minus signs combine, $-(-2) \cdot \frac{1}{2} = +1$, which is why the middle term changes from $-$ to $+$.

Step 4: Apply the limits

$$\begin{aligned} \text{RHS} &= \frac{1}{1+x^2} + \frac{1}{1+x^2} \left[\frac{1}{\sqrt{1+x^2}} - \frac{1}{\sqrt{1}} \right] = \cancel{\frac{1}{1+x^2}} + \frac{1}{(1+x^2)^{3/2}} - \cancel{\frac{1}{1+x^2}} \\ &= (1+x^2)^{-3/2} = y(x) = \text{LHS}. \end{aligned}$$

Conclusion

$y(x) = (1+x^2)^{-3/2}$ is a solution of the given Volterra I.E.

1.5 Problem 3: $y = xe^x$ **Problem 3**

Show that $y(x) = xe^x$ is a solution of the integral equation

$$y(x) = \sin x + 2 \int_0^x y(t) \cos(x-t) dt. \quad (3)$$

Correction to the notes

The statement in the manuscript reads $y(x) = xe^{-x}$, but the worked solution (and the true solution) is $y(x) = xe^x$. xe^{-x} **does not satisfy** (3). The statement has been corrected.

Independent check of the correction (Laplace transform)

Equation (3) is a **convolution** equation, $y = \sin x + 2(y * \cos)(x)$. With $Y = \mathcal{L}\{y\}$:

$$\begin{aligned} Y &= \frac{1}{s^2+1} + 2Y \cdot \frac{s}{s^2+1} \implies Y \frac{s^2-2s+1}{s^2+1} = \frac{1}{s^2+1} \\ \implies Y &= \frac{1}{(s-1)^2} \implies y(x) = xe^x. \quad \checkmark \end{aligned}$$

Step 1: Expand the kernel

Put $y(t) = te^t$ in the RHS of (3) and use $\cos(x-t) = \cos x \cos t + \sin x \sin t$:

$$\text{RHS} = \sin x + 2 \int_0^x te^t [\cos x \cos t + \sin x \sin t] dt.$$

Step 2: Split into two integrals

Since $\cos x$ and $\sin x$ are constants with respect to t ,

$$\text{RHS} = \sin x + 2 \cos x \underbrace{\int_0^x te^t \cos t dt}_{I_1} + 2 \sin x \underbrace{\int_0^x te^t \sin t dt}_{I_2}.$$

Standard results needed

$$\int e^t \cos t \, dt = \frac{e^t}{2}(\cos t + \sin t), \quad \int e^t \sin t \, dt = \frac{e^t}{2}(\sin t - \cos t).$$

Step 3: Evaluate I_1

Take $u = t$ ($u' = 1$) and $v = e^t \cos t$, with $v_1 = \frac{e^t}{2}(\cos t + \sin t)$. By parts,

$$I_1 = \left[\frac{t}{2} e^t (\cos t + \sin t) \right]_0^x - \int_0^x \frac{1}{2} e^t (\cos t + \sin t) \, dt.$$

The remaining integral is

$$\int \frac{1}{2} e^t (\cos t + \sin t) \, dt = \frac{1}{2} \left[\frac{e^t}{2} (\cos t + \sin t) + \frac{e^t}{2} (\sin t - \cos t) \right] = \frac{1}{2} \cdot \frac{e^t}{2} (2 \sin t) = \frac{1}{2} e^t \sin t.$$

Hence

$$I_1 = \left(\frac{x}{2} e^x (\cos x + \sin x) - 0 \right) - \left(\frac{1}{2} e^x \sin x - 0 \right).$$

 I_1

$$I_1 = \frac{x e^x}{2} (\cos x + \sin x) - \frac{e^x \sin x}{2}$$

Step 4: Evaluate I_2

Take $u = t$, $v = e^t \sin t$, with $v_1 = \frac{e^t}{2}(\sin t - \cos t)$:

$$I_2 = \left[\frac{t}{2} e^t (\sin t - \cos t) \right]_0^x - \int_0^x \frac{1}{2} e^t (\sin t - \cos t) \, dt.$$

The remaining integral is

$$\int \frac{1}{2} e^t (\sin t - \cos t) \, dt = \frac{1}{2} \left[\frac{e^t}{2} (\sin t - \cos t) - \frac{e^t}{2} (\cos t + \sin t) \right] = \frac{1}{2} \cdot \frac{e^t}{2} (-2 \cos t) = -\frac{1}{2} e^t \cos t.$$

Hence

$$I_2 = \left(\frac{x}{2} e^x (\sin x - \cos x) - 0 \right) - \left[-\frac{1}{2} e^t \cos t \right]_0^x = \frac{x}{2} e^x (\sin x - \cos x) + \left(\frac{1}{2} e^x \cos x - \frac{1}{2} \right).$$

 I_2

$$I_2 = \frac{x e^x}{2} (\sin x - \cos x) + \frac{e^x \cos x}{2} - \frac{1}{2}$$

Step 5: Assemble the RHS

$$\begin{aligned} \text{RHS} &= \sin x + 2 \cos x I_1 + 2 \sin x I_2 \\ &= \sin x + \cos x \left[x e^x (\cos x + \sin x) - e^x \sin x \right] + \sin x \left[x e^x (\sin x - \cos x) + e^x \cos x - 1 \right]. \end{aligned}$$

Expanding every product,

$$\begin{aligned} \text{RHS} &= \sin x + x e^x \cos^2 x + x e^x \sin x \cos x - e^x \sin x \cos x \\ &\quad + x e^x \sin^2 x - x e^x \sin x \cos x + e^x \sin x \cos x - \sin x. \end{aligned}$$

Step 6: Cancel and conclude

$$\begin{aligned}
\text{RHS} &= \cancel{\sin x} + xe^x \cos^2 x + \cancel{xe^x \sin x \cos x} - \cancel{e^x \sin x \cos x} \\
&\quad + xe^x \sin^2 x - \cancel{xe^x \sin x \cos x} + \cancel{e^x \sin x \cos x} - \cancel{\sin x} \\
&= xe^x [\cos^2 x + \sin^2 x] = xe^x = y(x) = \text{LHS}.
\end{aligned}$$

Conclusion

LHS = RHS, hence $y(x) = xe^x$ is a solution of the given integral equation.

1.6 Solved Exercises**1.6.1 Exercise 1: $y = 1 - x$** **Exercise 1 (from the notes)**

Show that $y(x) = 1 - x$ is a solution of the integral equation $\int_0^x e^{x-t} y(t) dt = x$.

Solution.

$$\text{LHS} = \int_0^x e^{x-t}(1-t) dt = e^x \int_0^x (1-t)e^{-t} dt.$$

Bernoulli's formula with $u = 1 - t$, $u' = -1$, $v = e^{-t}$, $v_1 = -e^{-t}$, $v_2 = e^{-t}$:

$$\begin{aligned}
\int (1-t)e^{-t} dt &= (1-t)(-e^{-t}) - (-1)(e^{-t}) = -(1-t)e^{-t} + e^{-t} = e^{-t}[t - 1 + 1] = te^{-t}, \\
\int_0^x (1-t)e^{-t} dt &= [te^{-t}]_0^x = xe^{-x} - 0 = xe^{-x}.
\end{aligned}$$

Conclusion

$$\text{LHS} = e^x \cdot xe^{-x} = x = \text{RHS} \implies y(x) = 1 - x \text{ is a solution}$$

1.6.2 Exercise 2: $y = 1$ **Exercise 2 (from the notes)**

Show that $y(x) = 1$ is a solution of the **Fredholm** integral equation

$$y(x) + \int_0^1 (e^{tx} - 1)xy(t) dt = e^x - x.$$

Solution. Put $y(t) = 1$ and $y(x) = 1$ in the LHS:

$$\text{LHS} = 1 + \int_0^1 (e^{tx} - 1)x dt = 1 + x \int_0^1 (e^{tx} - 1) dt.$$

Now

$$\int_0^1 e^{tx} dt = \left[\frac{e^{tx}}{x} \right]_0^1 = \frac{e^x - 1}{x} \quad (x \neq 0), \quad \int_0^1 1 dt = [t]_0^1 = 1,$$

so

$$\text{LHS} = 1 + x \left[\frac{e^x - 1}{x} - 1 \right] = 1 + \cancel{x} \cdot \frac{e^x - 1}{\cancel{x}} - x = 1 + e^x - 1 - x = e^x - x.$$

Conclusion

$\text{LHS} = e^x - x = \text{RHS} \implies \boxed{y(x) = 1 \text{ is a solution}}$

2 Method of Successive Approximations

2.1 The Iteration Scheme

Consider the Fredholm integral equation

$$y(x) = F(x) + \lambda \int_a^b K(x, t) y(t) dt \quad (4)$$

Iteration scheme

1. Take the **initial approximation** $y^{(0)}(x) = F(x)$.
2. Put $y(x) = y^{(0)}(x)$ in the RHS of (4) $\Rightarrow$ obtain $y^{(1)}(x)$.
3. Put $y(x) = y^{(1)}(x)$ in the RHS of (4) $\Rightarrow$ obtain $y^{(2)}(x)$.
4. Continue; after the n -th step we obtain $y^{(n)}(x) = F(x) + \lambda \int_a^b K(x, t) y^{(n-1)}(t) dt$.

The solution

$$y(x) = \lim_{n \rightarrow \infty} y^{(n)}(x)$$

Remarks on the scheme

- The same scheme applies verbatim to a **Volterra** equation — only the upper limit b is replaced by the variable x .
- The initial approximation need not be $F(x)$; any convenient starting function may be used, but $y^{(0)} = F$ is the standard choice.
- In practice one computes $y^{(1)}, y^{(2)}, y^{(3)}, \dots$, **recognises the pattern**, and sums the resulting series in closed form.

2.2 Problem 1: Fredholm Equation with Kernel xt

Problem 1

Solve, by the method of successive approximations, the integral equation

$$y(x) = 1 + \lambda \int_0^1 xt y(t) dt. \quad (5)$$

Observations. The limits 0 and 1 are constants, so this is a **Fredholm** equation with $F(x) = 1$ and (separable) kernel $K(x, t) = xt$. Initial approximation: $y^{(0)}(x) = 1$.

First approximation $y^{(1)}$

$$\begin{aligned}
 y^{(1)}(x) &= 1 + \lambda \int_0^1 x t (1) dt && \text{put } y(t) = y^{(0)}(t) = 1 \\
 &= 1 + \lambda x \int_0^1 t dt && x \text{ is constant w.r.t. } t \\
 &= 1 + \lambda x \left[\frac{t^2}{2} \right]_0^1 = 1 + \lambda x \left[\frac{1}{2} - 0 \right] = 1 + \frac{\lambda x}{2}.
 \end{aligned}$$

Second approximation $y^{(2)}$

Substitute $y(t) = y^{(1)}(t) = 1 + \frac{\lambda t}{2}$ in (5):

$$\begin{aligned}
 y^{(2)}(x) &= 1 + \lambda \int_0^1 x t \left(1 + \frac{\lambda t}{2} \right) dt = 1 + \lambda x \int_0^1 \left(t + \frac{\lambda t^2}{2} \right) dt \\
 &= 1 + \lambda x \left[\left(\frac{t^2}{2} \right)_0^1 + \lambda \left(\frac{t^3}{6} \right)_0^1 \right] = 1 + \lambda x \left[\frac{1}{2} + \frac{\lambda}{6} \right].
 \end{aligned}$$

 $y^{(2)}$

$$y^{(2)}(x) = 1 + \frac{\lambda x}{2} + \frac{\lambda^2 x}{6} = 1 + \frac{\lambda x}{2} \left[1 + \frac{\lambda}{3} \right], \quad \text{since } \frac{\lambda^2 x}{6} = \frac{\lambda x}{2} \cdot \frac{\lambda}{3}.$$

Third approximation $y^{(3)}$

Substitute $y(t) = y^{(2)}(t) = 1 + \frac{\lambda t}{2} + \frac{\lambda^2 t}{6}$ in (5):

$$y^{(3)}(x) = 1 + \lambda \int_0^1 x t \left[1 + \frac{\lambda t}{2} + \frac{\lambda^2 t}{6} \right] dt = 1 + \lambda x \int_0^1 \left(t + \frac{\lambda t^2}{2} + \frac{\lambda^2 t^2}{6} \right) dt.$$

Careful!

$y^{(2)}(t)$ is **linear** in t — the last term is $\frac{\lambda^2 t}{6}$, not $\frac{\lambda^2 t^2}{6}$. Multiplying by t therefore gives t^2 , not t^3 .

$$\begin{aligned}
 y^{(3)}(x) &= 1 + \lambda x \left[\left(\frac{t^2}{2} \right)_0^1 + \lambda \left(\frac{t^3}{6} \right)_0^1 + \frac{\lambda^2}{6} \left(\frac{t^3}{3} \right)_0^1 \right] = 1 + \lambda x \left[\frac{1}{2} + \frac{\lambda}{6} + \frac{\lambda^2}{18} \right] \\
 &= 1 + \frac{\lambda x}{2} + \frac{\lambda^2 x}{6} + \frac{\lambda^3 x}{18}.
 \end{aligned}$$

 $y^{(3)}$, factorised

$$y^{(3)}(x) = 1 + \frac{\lambda x}{2} \left[1 + \left(\frac{\lambda}{3} \right) + \left(\frac{\lambda}{3} \right)^2 \right]$$

The pattern

Collecting the approximations

$$\begin{aligned}
y^{(0)}(x) &= 1 \\
y^{(1)}(x) &= 1 + \frac{\lambda x}{2} \\
y^{(2)}(x) &= 1 + \frac{\lambda x}{2} \left[1 + \left(\frac{\lambda}{3} \right) \right] \\
y^{(3)}(x) &= 1 + \frac{\lambda x}{2} \left[1 + \left(\frac{\lambda}{3} \right) + \left(\frac{\lambda}{3} \right)^2 \right] \\
y^{(4)}(x) &= 1 + \frac{\lambda x}{2} \left[1 + \left(\frac{\lambda}{3} \right) + \left(\frac{\lambda}{3} \right)^2 + \left(\frac{\lambda}{3} \right)^3 \right]
\end{aligned}$$

General term

$$y^{(n)}(x) = 1 + \frac{\lambda x}{2} \left[1 + \left(\frac{\lambda}{3} \right) + \left(\frac{\lambda}{3} \right)^2 + \cdots + \left(\frac{\lambda}{3} \right)^{n-1} \right]$$

Passing to the limit and summing the series

$$y(x) = \lim_{n \rightarrow \infty} y^{(n)}(x) = 1 + \frac{\lambda x}{2} \left[1 + \left(\frac{\lambda}{3} \right) + \left(\frac{\lambda}{3} \right)^2 + \left(\frac{\lambda}{3} \right)^3 + \cdots \right].$$

Using the geometric (binomial) series $(1 - u)^{-1} = 1 + u + u^2 + u^3 + \cdots$ for $|u| < 1$, with $u = \lambda/3$:

$$y(x) = 1 + \frac{\lambda x}{2} \left[1 - \frac{\lambda}{3} \right]^{-1} = 1 + \frac{\lambda x}{2} \left[\frac{3 - \lambda}{3} \right]^{-1} = 1 + \frac{\lambda x}{2} \left(\frac{3}{3 - \lambda} \right).$$

Required solution

$$y(x) = 1 + \frac{3\lambda x}{2(3 - \lambda)}, \quad |\lambda| < 3.$$

Verification by the separable-kernel method

Since $K = xt$, write $y(x) = 1 + \lambda x C$ with $C = \int_0^1 t y(t) dt$. Then

$$\begin{aligned}
C &= \int_0^1 t(1 + \lambda t C) dt = \left[\frac{t^2}{2} \right]_0^1 + \lambda C \left[\frac{t^3}{3} \right]_0^1 = \frac{1}{2} + \frac{\lambda C}{3}, \\
\Rightarrow C \left(1 - \frac{\lambda}{3} \right) &= \frac{1}{2} \Rightarrow C = \frac{3}{2(3 - \lambda)},
\end{aligned}$$

$$\text{so } y(x) = 1 + \lambda x \cdot \frac{3}{2(3 - \lambda)} = 1 + \frac{3\lambda x}{2(3 - \lambda)} \quad \checkmark$$

2.3 Problem 2: Volterra Equation with Kernel $x - t$ **Problem 2**

Using the method of successive approximations, solve the **Volterra** integral equation

$$y(x) = 1 + x + \int_0^x (x - t) y(t) dt. \quad (6)$$

Setup. $F(x) = 1 + x$, kernel $K(x, t) = x - t$, variable upper limit $\Rightarrow$ Volterra. Initial approximation: $y^{(0)}(x) = 1 + x$.

First approximation $y^{(1)}$

Substitute $y(t) = y^{(0)}(t) = 1 + t$ in (6). Since $(x - t)(1 + t) = x + xt - t - t^2 = x(1 + t) - (t + t^2)$,

$$\begin{aligned} y^{(1)}(x) &= 1 + x + \int_0^x (x - t)(1 + t) dt = 1 + x + x \int_0^x (1 + t) dt - \int_0^x (t + t^2) dt \\ &= 1 + x + x \left[t + \frac{t^2}{2} \right]_0^x - \left[\frac{t^2}{2} + \frac{t^3}{3} \right]_0^x = 1 + x + x \left(x + \frac{x^2}{2} \right) - \left(\frac{x^2}{2} + \frac{x^3}{3} \right) \\ &= 1 + x + x^2 + \frac{x^3}{2} - \frac{x^2}{2} - \frac{x^3}{3} = 1 + x + \frac{x^2}{2} + x^3 \left(\frac{1}{2} - \frac{1}{3} \right). \end{aligned}$$

Since $\frac{1}{2} - \frac{1}{3} = \frac{3-2}{6} = \frac{1}{6}$:

$y^{(1)}$

$$y^{(1)}(x) = 1 + x + \frac{x^2}{2} + \frac{x^3}{6} = 1 + \frac{x}{1!} + \frac{x^2}{2!} + \frac{x^3}{3!}$$

These are already the first four terms of the Maclaurin series of e^x .

Second approximation $y^{(2)}$

Substitute $y(t) = y^{(1)}(t) = 1 + t + \frac{t^2}{2} + \frac{t^3}{6}$ in (6):

$$\begin{aligned} y^{(2)}(x) &= 1 + x + \int_0^x (x - t) \left(1 + t + \frac{t^2}{2} + \frac{t^3}{6} \right) dt \\ &= (1 + x) + x \int_0^x \left(1 + t + \frac{t^2}{2} + \frac{t^3}{6} \right) dt - \int_0^x \left(t + t^2 + \frac{t^3}{2} + \frac{t^4}{6} \right) dt \\ &= 1 + x + x \left[t + \frac{t^2}{2} + \frac{t^3}{6} + \frac{t^4}{24} \right]_0^x - \left[\frac{t^2}{2} + \frac{t^3}{3} + \frac{t^4}{8} + \frac{t^5}{30} \right]_0^x \\ &= 1 + x + x^2 + \frac{x^3}{2} + \frac{x^4}{6} + \frac{x^5}{24} - \frac{x^2}{2} - \frac{x^3}{3} - \frac{x^4}{8} - \frac{x^5}{30} \\ &= 1 + x + \left(1 - \frac{1}{2} \right) x^2 + \left(\frac{1}{2} - \frac{1}{3} \right) x^3 + \left(\frac{1}{6} - \frac{1}{8} \right) x^4 + \left(\frac{1}{24} - \frac{1}{30} \right) x^5. \end{aligned}$$

Arithmetic of the coefficients

$$1 - \frac{1}{2} = \frac{1}{2}, \quad \frac{1}{2} - \frac{1}{3} = \frac{1}{6}, \quad \frac{1}{6} - \frac{1}{8} = \frac{4-3}{24} = \frac{1}{24}, \quad \frac{1}{24} - \frac{1}{30} = \frac{5-4}{120} = \frac{1}{120}$$

$y^{(2)}$

$$y^{(2)}(x) = 1 + x + \frac{x^2}{2} + \frac{x^3}{6} + \frac{x^4}{24} + \frac{x^5}{120} = 1 + \frac{x}{1!} + \frac{x^2}{2!} + \frac{x^3}{3!} + \frac{x^4}{4!} + \frac{x^5}{5!}$$

Each iteration adds **two** more terms of the exponential series.

The pattern and the limit

So far

$$y^{(0)}(x) = 1 + x$$

$$y^{(1)}(x) = 1 + \frac{x}{1!} + \frac{x^2}{2!} + \frac{x^3}{3!}$$

$$y^{(2)}(x) = 1 + \frac{x}{1!} + \frac{x^2}{2!} + \frac{x^3}{3!} + \frac{x^4}{4!} + \frac{x^5}{5!}$$

$$y^{(3)}(x) = 1 + \frac{x}{1!} + \frac{x^2}{2!} + \cdots + \frac{x^6}{6!} + \frac{x^7}{7!}$$

General term: $y^{(n)}(x) = \sum_{k=0}^{2n+1} \frac{x^k}{k!}.$

$$y(x) = \lim_{n \rightarrow \infty} y^{(n)}(x) = 1 + \frac{x}{1!} + \frac{x^2}{2!} + \frac{x^3}{3!} + \frac{x^4}{4!} + \cdots = \sum_{k=0}^{\infty} \frac{x^k}{k!}.$$

Required solution

$$y(x) = e^x$$

Verification: reduce (6) to an initial value problem

Differentiate $y(x) = 1 + x + \int_0^x (x-t)y(t) dt$ using Leibniz's rule:

$$y'(x) = 1 + \underbrace{(x-t)y(t)}_{=0} \Big|_{t=x} + \int_0^x \frac{\partial}{\partial x} (x-t) y(t) dt = 1 + \int_0^x y(t) dt,$$

$$y''(x) = y(x).$$

With $y(0) = 1$ and $y'(0) = 1$, the solution of $y'' = y$ is $y = Ae^x + Be^{-x}$ with $A = 1$, $B = 0$, i.e. $y(x) = e^x$ ✓

2.4 Exercise Set

Solve by the method of successive approximations

(1) $y(x) = x + \int_0^x (t-x)y(t) dt$

$$(2) \quad y(x) = 1 - \lambda \int_0^1 x t y(t) dt$$

$$(3) \quad y(x) = 1 + \lambda \int_0^1 (1 - 3xt)y(t) dt$$

$$(4) \quad y(x) = 2(1 + x^2) - \int_0^x x y(t) dt$$

$$(5) \quad y(x) = 1 + x - \int_0^x y(t) dt$$

How to attack them

Separable kernels, (2) and (3): write y in terms of the unknown constants $\int_0^1 y(t) dt$ and $\int_0^1 t y(t) dt$, then solve the resulting linear system.

Volterra kernels, (1), (4) and (5): either iterate as in Problem 2, or differentiate to convert the equation into an initial value problem: (1) $y'' = -y$, $y(0) = 0$, $y'(0) = 1$; (5) $y' = 1 - y$, $y(0) = 1$.

Answers for self-checking

$$(1) \quad y(x) = \sin x \quad (\text{from } y'' = -y, y(0) = 0, y'(0) = 1)$$

$$(2) \quad y(x) = 1 - \frac{3\lambda x}{2(3 + \lambda)}$$

$$(3) \quad y(x) = \frac{4 + 4\lambda - 6\lambda x}{4 - \lambda^2} \quad (\lambda \neq \pm 2)$$

$$(4) \quad y(x) = 2 \quad (\text{check: } 2(1 + x^2) - 2x^2 = 2)$$

$$(5) \quad y(x) = 1 \quad (\text{check: } 1 + x - x = 1)$$

3 Corrections and Summary

3.1 Corrections Made to the Manuscript

List of corrections

- Section 1, Problem 3.** Statement written as $y(x) = xe^{-x}$; corrected to $y(x) = xe^x$, which is what the worked solution uses and what actually satisfies the equation (confirmed by Laplace transform).
- Section 1, Problem 1.** The limit substitution $[-(2-t)e^{-t}]_0^x$ contributes +2 at the lower limit; every intermediate line is now written out explicitly.
- Section 2, Problem 1.** Emphasised that $y^{(2)}(t)$ is **linear** in t , so $t \cdot y^{(2)}(t)$ produces t^2 (not t^3) — this is where the coefficient $\lambda^2/18$ comes from.
- Series summation restricted to $|\lambda| < 3$ in Section 2, Problem 1.

3.2 Summary

Verification of solutions

- Substitute the candidate y , evaluate the integral fully, compare sides.
- Tools used: Bernoulli's (generalised parts) formula, the substitution $m = 1 + t^2$, expansion of $\cos(x - t)$.

Method of successive approximations

- $y^{(0)} = F$, then $y^{(n)} = F + \lambda \int K y^{(n-1)} dt$, and $y = \lim_{n \rightarrow \infty} y^{(n)}$.
- Fredholm, $K = xt$: geometric series $\Rightarrow y = 1 + \frac{3\lambda x}{2(3 - \lambda)}$, $|\lambda| < 3$.
- Volterra, $K = x - t$: exponential series $\Rightarrow y = e^x$.